

Research

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Research topics

Keywords : algorithm design, parameterized complexity, scheduling, graph theory.

My research mainly consists in designing efficient algorithms for scheduling problems and studying the computational complexity of these problems.

Motivation

Most scheduling problems are strongly NP-hard and have required intricate heuristics to be solved in practice. While yielding results short term, such heuristics rarely help to identify what fundamentally makes these problems difficult. Such analysis would uncover the similarities between seemingly different problems and facilitate strategy transfers. This would also prove useful whenever one would like to enrich a problem - for example by adding precedence relations or going from equal-length tasks to tasks of arbitrary time length.

One way to gather such knowledge is to go beyond classical complexity theory and consider parameterized complexity theory. Given a problem $\mathcal{P}$ we choose parameter k as some property of the input like the number of machines or the width of the precedence graph (if there is one). Depending on problem $\mathcal{P}$ and parameter k either we design algorithms which are *fixed-parameter tractable* with respect to k (i.e. which operate in time $f(k) \times \text{poly}(n)$ with f an arbitrary computable function), or we show that $\mathcal{P}$ remains hard even when k is small (typically via a reduction from a well-known difficult parameterized problem like k -CLIQUE or k -COLORING).

Then the set of parameters for which problem $\mathcal{P}$ is *fixed-parameter tractable* can be interpreted as a footprint. By comparing it to the set of working parameters from another NP-hard problem $\mathcal{P}'$, one could infer whether $\mathcal{P}$ is 'equivalent' to $\mathcal{P}'$, 'strictly harder' than $\mathcal{P}'$, or if both problems are difficult for different reasons.

Research activities

Currently : postdoctoral researcher at LIP, École Normale Supérieure de Lyon Lyon, France

— Planned duration : eighteen months (i.e., until April 2026).

Nov. 2024 - Now

— *Subject : Scheduling radio-astronomical observations*

During this postdoctoral research project supervised by Frédéric Vivien and Anne Benoit, we study the scheduling of radio-astronomical observations. Because the Earth revolves around itself, for a ground-based observer the height of a celestial body in the sky changes over time periodically with a (roughly) twenty-four hour long period. Although one can set a minimum altitude threshold to mitigate the impact of Earth's atmosphere on observation quality, this defines a daily availability window for each celestial body. While scheduling problems featuring collections of availability windows have historically been difficult to optimize and even to approximate, the periodicity of these time windows could play a role in making the problem easier.

This work is done in collaboration with astrophysicists from the ECLAT laboratory, who have shared scheduling data from the NenuFAR radio-telescope over several weeks. During the first few months of this project, we studied the computational complexity - both classic and parameterized - of a non-preemptive model of the problem. This has led to the design of several algorithms - both exact and approximation algorithms - and heuristics. These procedures are currently being implemented and confronted to the NenuFAR data. In the near future, meetings with ECLAT members are planned to discuss the model and find the best compromise between observation quality and makespan minimization.

— Duration : three years.

Oct. 2021 - Oct. 2024

— *Subject : Parameterized complexity and new efficient enumerative schemes for RCPSP*

During my Ph.D. thesis supervised by Claire Hanen, I tackled the inclusion of job time windows and precedence delays to RCPSP and its subproblems. With precedence delays we considered parameter ℓ_{max} - i.e. the maximum delay value appearing in the input - in the case of minimum delays, exact delays and the lesser studied maximum delays. For all delay types it was shown that scheduling unit-time jobs on a single machine with a single available delay value is hard even with small ℓ_{max} . In the case of minimum delays we also showed that the parameterized problem remains hard with chains and two distinct available delay values. This suggests that another property of the problem has to be bounded in order to deal with precedence delays.

With job time windows we considered several parameters describing their structure. Notably *slack* σ - i.e. the maximum difference between the time window length of a job and its processing time -, and *pathwidth* μ - i.e. the maximum number of job time windows which can include the same time unit - which have shown recent success in the literature. We showed that scheduling unit-time coupled tasks on a single machine with time windows is para-NP-hard with respect to either parameter.

While this rules out any hope for a *fixed-parameter tractable* algorithm with precedence delays and these parameters alone, the reduction techniques that we used hint at the underlying reason. Indeed unbounded delay values are used to circumvent the job time window restrictions put by these parameters. And fittingly we showed that the problem becomes *fixed-parameter tractable* if you combine them with maximum delay value ℓ_{max} . In fact we can even have general precedence with any number of distinct delay values, either on parallel machines or with arbitrary processing times. With multiple machines and arbitrary processing times the problem becomes hard again, and you would have to add an extra parameter - like the maximum processing time - to get a *fixed-parameter tractable* algorithm.

Finally with job time windows and no precedence delays we introduced a new parameter called the *proper level* pl . Parameter pl is defined as the maximum number of job time windows which can strictly include - on both ends - another job time window. On single machine scheduling with job time windows we proposed a new dominance rule which generalizes the *Earliest Deadline* rule and leads to a *fixed-parameter tractable* number of dominant schedules with respect to pl . The resulting makespan algorithm is quasi-linear in the input size with a single exponential dependency on pl , which gives it a good chance to be competitive in future applications.